CHAPTER NINE

PRICE RELATIONSHIPS AND PRICE CHANGES IN SELECTED AGRICULTURAL MARKETS

SEBASTIAN HESS AND ULRICH KOESTER

9.1. Introduction and learning objectives

This chapter examines selected relationships between individual agricultural prices. Terms that are sometimes used more or less synonymously in this context are the **agricultural price structure, price relations** or **price ratios.**

In Chapter 2 we derived various optimal equilibrium price relationships between outputs and inputs under neoclassical conditions. We seldom observe these price relationships in reality, however, because the assumptions of neoclassical economics are often violated. In addition, agricultural markets are seldom in equilibrium. Instead, markets are typically adjusting to shocks, for example because market participants have received new information, due to changes in behaviour, preferences and technology, and in response to changes in institutions such as government regulations. As a result of such shocks, equilibria are disturbed, which triggers market processes that change prices and quantities in the direction of new equilibria. Since markets are interrelated, adjustments on one market will inevitably lead to adjustments on others.

In this chapter we will analyse the reactions of farmers and processors of agricultural products to selected changes in agricultural prices. As we will see, changes in price relations on agricultural markets are often delayed compared with other sectors. Adjustments are delayed by specific conditions in agriculture, such as biological production processes that take time to unfold (think of tree crops, or the time it takes to expand a dairy herd), the limited availability of land, and the fact that agriculture often produces joint products (such as protein and fat in milk, or oil and meal in oilseeds). But institutions such as laws and the culturally determined behaviour of farmers can also slow down market adjustments in agriculture. It is not the aim of this chapter to derive exactly which price relations will arise in reality as a result of certain shocks. Instead, the primary aim is to analyse and explain how price relationships can be expected to change in response to selected shocks.

Changes in agricultural price relations are not only caused by changes in economic incentive structures but also by institutions. Agriculture produces both private and public goods, and both types of goods are important for the achievement of agricultural policy goals. It is therefore understandable that governments often intervene directly or indirectly on agricultural markets through a large variety of measures (laws and regulations, i.e. institutions) that influence price relations. Laws and regulations that affect agricultural price relationships are not only implemented by national governments, but also at the supra-national level, for example by the EU or the World Trade Organisation (WTO).

In addition, agricultural markets are integrated in increasingly complex value-added chains. Value is added to an agricultural raw product at each link in the chain, for example as milk is processed into milk powder, then baby formula, etc. Hence, price relationships in agriculture are not only horizontal (for example between wheat and barley, or milk and beef), but also vertical (for example between milk and butter, or wheat and bread).

In this chapter we will:

- analyse horizontal price relationships between agricultural products using selected examples,
- discuss price relationships between agricultural raw products and the joint products that they are processed into (for example between milk, milk fat and protein),
- provide an overview of how institutions can affect agricultural price relations using several important examples from the history of agricultural policy in the EU.

9.2. Price relationships between agricultural goods at the same processing stage

Neoclassical models of agricultural pricing usually assume that goods are homogeneous, i.e. different qualities of a good such as wheat are treated as separate goods. In this section, therefore, we will first discuss price determination for homogeneous agricultural goods at the same processing stage, before considering heterogeneous goods. After that we will present examples that illustrate how the effectiveness of policy measures on a particular market are affected by interactions with other markets.

9.2.1. Equilibrium price relationships for related agricultural goods

Price changes between two related agricultural markets can first be examined using a simple, comparative-static partial equilibrium model.[1] Consider two interrelated markets in a closed economy, i.e. one that does not trade with other countries. Without loss of generality we refer to the two markets as 'poultry meat' (M) and 'grain' (G). The supply and demand curves for these markets are:

poultry meat supply: $q_M^S = A\, p_M^a\, p_G^{-b}$ (1)

poultry meat demand: $q_M^D = \ B\, p_M^{-d}$ (2)

grain supply: $q_G^S = C\, p_G^e$ (3)

grain demand: $q_G^D = E\, p_G^{-g}\, p_M^h$ (4)

where the constants A, B, C and E are all greater than zero, as are a, b, d, e, g and h.

In equation (1) the market supply for poultry meat is an increasing function of the meat price, and a decreasing function of the grain price, the latter because grains are an important feed input in poultry meat production. Rising grain prices thus increase the marginal costs of meat production; *ceteris paribus* this leads to a reduction in the supply of meat. The supply of grain in equation (3) is an (increasing) function of the grain price alone, and the demand for meat in equation (2) is a (decreasing) function of the meat price alone. However, the demand for grain in equation (4) depends on both the grain price and the meat price. An increase in the meat price increases the demand for grain by way of an expansion in meat production.

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[1] See Roningen (1997, pp. 231-257) and Francois and Hall (1997, pp. 122-155).

We solve for equilibrium prices by equating supply and demand:

Equilibrium condition:

$$\mathrm{A}\, p_M^a\, p_G^{-b} = \mathrm{B}\, p_M^{-d} \text{ and } C p_G^e = \; E p_G^{-g}\, p_M^h \tag{5}$$

Equilibrium prices:

$$p_M^* = \left(\frac{B}{A}\right)^{\left(\frac{1}{a+d}\right)} p_G^{\left(\frac{b}{a+d}\right)} \text{ and } p_G^* = \left(\frac{E}{C}\right)^{\left(\frac{1}{e+g}\right)} p_M^{\left(\frac{h}{e+g}\right)} \tag{6}$$

We see that the equilibrium price of poultry meat is a function of the grain price, and *vice versa*. The interactions between these prices are determined by the relative magnitudes of the various supply and demand elasticities.

The price of grain has always played a particularly important role in the history of agricultural price and market policy. This price is sometimes referred to as **pivotal agricultural price** because of grain's importance for human nutrition, as a feedstuff in livestock production, and also as feedstock in energy production (e.g. when maize is processed into ethanol). Hence, the grain price has a similar effect on the overall agricultural price structure as the basic wage rate has on the wage structure in an economy.[2]

The model in equations (1) through (6) shows how a change in the grain price affects other agricultural prices and thus the overall agricultural price structure. In reality, of course, more than two markets and prices will interact. The resulting price effects will, as a rule, be non-linear and depend on the relative magnitudes of the supply and demand elasticities involved. As a result, a shock on one market can trigger disproportionately smaller or larger price reactions on other markets.

9.2.2. The supply of and demand for inputs

The supply and demand functions in equations (1) through (4) above depict the aggregated behaviour of many individual market participants (producers and consumers). On the supply side, for example, each individual poultry producer contributes to aggregate supply based on the size of his/her operation and the specific production technology used. This technology

[2] In some countries where wages are set by collective bargaining agreements between groups of employers and large unions, the agreement reached in a particular sector will define a basic wage rate that serves as a benchmark for negotiations in other sectors.

describes how inputs such as feed are combined and transformed into an output (poultry meat). To illustrate this let us assume that poultry meat production follows the following Cobb-Douglas production technology: $q_M = q_G{}^{0,5} q_S{}^{0,5}$.

According to this technology, poultry meat (q_M) is produced using the inputs grain (q_G) and soymeal (q_S). The exponents on the input quantities are **production elasticities** that indicate by how many percentage points the output of poultry meat will increase if the respective input amount is increased by 1%. Since the assumed production technology is characterised by constant returns to scale (the sum of the production elasticities equals 1) the exponents can also be interpreted as cost shares. Hence, grain and soymeal each account for 50% of the costs of production.

The indirect (or **dual**)[3] Cobb-Douglas **cost function** corresponding to this production technology is:

$$cost^{min} = 2q_M(p_G)^{0,5}(p_S)^{0,5} \tag{7}$$

The derivation of this function assumes that the producer always aims to produce a given quantity of poultry meat (q_M) at minimum cost ($cost^{min}$) given the prices of grain (p_G) and soymeal (p_S).

Making use of **Shephard's Lemma**, differentiating equation (7) with respect to the two input prices leads to the (Hicksian or compensated) input demand functions. These functions describe poultry meat producers' behaviour on the markets for grain and for soymeal. The resulting input demand for grain is:

$$\frac{\partial c^{min}}{\partial p_G} = q_M(p_G)^{-0.5}(p_S)^{0.5} = q_G \tag{8}$$

and the corresponding input demand for soymeal is:

$$\frac{\partial c^{min}}{\partial p_S} = q_M(p_G)^{0.5}(p_S)^{-0.5} = q_S \tag{9}$$

The demand functions in equations (8) and (9) can be linearized by taking logarithms. This makes it possible to estimate the production elasticities directly using regression techniques and subsequently to test hypotheses about the magnitudes of these elasticities. For the example of the input

[3] Snyder et al. (2015) explain microeconomic duality theory.

demand for grain in equation (8), equation (10) shows the log-linear regression model that can be estimated using a sample of observations on $i = 1, 2, \ldots N$ poultry meat producers:

$$ln(q_G)_i = \hat{\beta}_0 + \hat{\beta}_G ln(p_G)_i + \hat{\beta}_S ln(p_S)_i + \hat{u}_i \qquad (10)$$

Estimation of equation (10) can be used to test hypotheses such as $\beta_G = -0.5$ and $\beta_S = 0.5$ about the individual production elasticities, or $|\beta_G| + \beta_S = 1$ (i.e. whether the production technology is characterised by constant returns to scale). In equation (10) the coefficients β_G and β_S can also be interpreted as price elasticities of factor demand.

Table 9.1 presents estimated price elasticities of demand for different feedstuffs that have been econometrically estimated for the EU. We see that the demand for the cereal substitute tapioca is especially price elastic, as a 1% increase in the price of tapioca in the EU (for example as a result of an increase in the world market price) leads to a 2.1% reduction in demand.

Table 9.1: Price elasticities of demand for different feedstuffs in the EU (1979-1998).

	A 1% increase in the price of…			
…leads to an X% change in the demand for	Wheat	Other cereals	Protein	Tapioca
Wheat	-1.212	1.412	-0.160	-0.040
Other cereals	0.541	-0.704	0.137	0.026
Protein	-0.105	0.233	-0.323	0.194
Tapioca	-0.264	0.440	1.956	-2.132

Source: Rude and Meilke (2000).

The cost function in equation (7) can also be used to derive the marginal costs of poultry meat production and thus the market supply. For this purpose (7) is differentiated with respect to q_M which results in:

$$\frac{\partial c^{min}}{\partial q_M} = 2(p_G)^{0.5}(p_S)^{0.5} \qquad (11)$$

9.2.3. Agricultural goods with heterogeneous qualities: Hedonic prices

In the previous sections we analysed price relationships between products such as poultry meat and grain that we assumed to be homogeneous. However, it is also possible to analyse price relationships between different qualities or types of the same product, for example between different vintages and varieties of wine.[4] In this case, the product in question is heterogeneous. Often, the market prices for agricultural products that are published, for example by government statistical offices, are averages of prices from a large number of individual transactions over a specific period of time. Of course, not every piglet or ton of barley that was traded in a given week, for example, changed hands at the exact price shown in the respective weekly price quotation. If the underlying distribution of individual transaction prices is skewed, then the average transaction price will not equal the most frequent (modal) transaction price. If the underlying price distribution is multi-modal, then the average price might not provide any meaningful information at all. For this reason, it can be instructive to take a closer look at the distribution of prices for heterogeneous good, as we illustrate in the following for the example of horses.[5]

At the end of an auction it is possible to calculate the average price paid over all of the horses sold. However, the price of a particular horse sold at an auction will not depend only, and perhaps not at all, on the total number of horses sold and the total number of buyers present. Instead, the price of a particular horse $k \in \{1, 2, \ldots, K\}$ will depend on a vector of its performance characteristics and other factors. If there are n such characteristics, then horse k can be described by the vector $Z_k = z_{1k}, z_{2k}, \ldots, z_{nk}$, and the price of that horse is given by:

$$p_k(Z_k) = p(z_{1k}, z_{2k}, \ldots, z_{nk}) \tag{12}$$

A seller will not only offer horse k but will also sell it if the auction price $p_k(Z_k)$ equals the marginal costs $MC(Z_k)$ of raising a horse with the characteristics Z_k. Hence, the number of horses sold at an auction, Q^* depends on the prices $p_k(Z_k)$ which in turn depend on the characteristics of the individual horses:

[4] See for example Schäufele et al. (2016).
[5] See for example Stowe (2013) or Hess et al. (2014). The implications of product differentiation for (imperfect) competition are analysed in detail in Chapter 8.

$$Q^{k*} = Q^{k*}(P, Z_1, Z_2, \dots, Z_k) \tag{13}$$

On the demand side, we assume that the different performance characteristics in the vector Z generate benefits for the buyer according to a utility function: $U = U\ (z_1, z_2, \dots,\ z_n, x)$, where x is a vector that includes all other relevant factors that are not specific to a particular horse (such as the distance of the auction from the buyer's home, and any fees that must be paid to participate in the auction). Horse buyers maximise their utility by demanding horses with certain characteristics, subject to their budget constraints:

$$I = p_x x + p(z_1, z_2, \dots, z_n) \tag{14}$$

A buyer's expenditure for a certain desired characteristic thus depends on the horse's other characteristics and their prices, as well as the cost of any other expenses as captured by x.

On a market with many suppliers of horses and many buyers, there will be a supply $S_z(z_n)$ and a demand $D_z(z_n)$ for each individual characteristic z_n, and together these will determine the equilibrium amount of this characteristic $S_z^*(z_n) = D_z^*(z_n)$ and its equilibrium price $p_z^*(z_n)$. These prices for individual characteristics are rarely observed, but they are aggregated over all characteristics for each individual horse leading to its specific price $p_k(Z_k)$. For example, in an auction of race horses one does not explicitly observe how much the price of a horse increases for each additional 1000 Euro of prize money it has won to date. However, given sufficient information on the characteristics and sales prices of a sufficiently large number of horses, it is possible to calculate the implicit value of an additional 1000 Euro of prize winnings, using what is referred to as the **hedonic price model**. The hedonic price model explains the variation in prices among a set of heterogeneous goods within a product category (such as horses) as a function of the goods' characteristics and the equilibrium prices of these characteristics:

$$P_k^*(Z_k) = f(z_{1k}^*, z_{2k}^*, \dots, z_{nk}^*) \tag{15}$$

Hedonic price models are of great practical importance in empirical market analysis. For estimation purposes, a stochastic error term is added to the equation (15), which is then estimated using regression techniques:

$$P_k^*(Z_k) = \beta_0 + \beta_1 z_{1k}^* + \beta_2 z_{2k}^* + \cdots + \beta_n z_{nk}^* + u_k \tag{16}$$

For the example of a horse auction, the $p_k^*(Z_k)$ are the observed auction prices for a sample of horses. On the right side of the equation, the explanatory variables z_{1k}^* to z_{nk}^* are the relevant characteristics of each horse, such as its height, gender, colour, and jumping ability, etc. The regression coefficients β measure the marginal increase in the auction price for an average horse if the corresponding characteristic increased by one unit. Hedonic price models are thus used to estimate the market mark-ups (or mark-downs) that correspond to specific product characteristics. Hedonic analysis has been used in the economic literature for a wide variety of heterogenous products such as wine, real estate, and agricultural land, and it often underlies online market portals that provide value estimates for products such as used vehicles and homes.

9.2.4. The practical importance of substitutability on agricultural markets

Price relationships also arise between goods at the same processing stage when they are substitutes for one another. In the following we discuss this using the example of grain and grain substitutes in animal feed that was touched upon above (see Table 9.1). The competitiveness of a particular feedstuff depends on its nutrient content and its price relative to those of other feedstuffs.

In practice, linear programming (LP) models are often used to optimise feed rations. In addition, a method proposed by Löhr for evaluating the value of a feedstuff can be used. This method considers two nutrients, for example protein and energy. For dairy cattle, the energy content of feed is often measured in MJ NEL (megajoule net energy content for lactation), which indicates the energy content in joules that can be metabolised for milk production per kg of the feed. Protein is measured according to the raw protein content. **Löhr's method** can be used, for example, to calculate how much wheat and soybean meal are required to replace one kg of another feed. Alternatively, the method can be used to calculate up to what price a substitute feed can compete with other feeds such as wheat and soybean meal.

To illustrate Löhr's method we use information about the raw protein and energy content as well as the prices of wheat, soybean meal, and the potential substitute feed. This is demonstrated in the following example in which we calculate the prices at which rapeseed meal and other feedstuffs can compete with a mixture of soybean meal and wheat in dairy rations.

First, the necessary information on protein and energy contents and prices for wheat and soybean meal are presented in Table 9.2.

Table 9.2: Comparative feed levels in a dairy cattle ration.

Alternative feed	Raw protein in g / kg	MJ NEL / kg dry matter	Market prices in € / dt
Soybean meal	425	7,59	28.00
Wheat	121	7,49	14.00

Source: Own depiction based on Groß (2010).

Based on this information, Löhr's method begins by solving the following system of equations to calculated the implicit prices of raw protein (x) and energy (y) in wheat and soybean meal:

Soybean meal: $425x + 7.59y = 28$ (17)

Wheat: $121x + 7.49y = 14$ (18)

The resulting values are

$y = 1.1312$ (the implicit price of energy), and

$x = 0.0456$ (the implicit price of raw protein).

With the help of these implicit prices, we next calculate maximum price at which an alternative feedstuff can compete with a mixture of wheat and soybean meal. One kg of rapeseed meal, for example, contains 330 g of raw protein and 6.51 MJ NEL (see Table 9.3). A reference price (p^{ref}) for rapeseed meal is calculated by multiplying these amounts with their respective implicit prices:

$$p^{ref} = 330 \cdot 0{,}0456 + 6{,}51 \cdot 1{,}1312 = 22{,}44 \quad (19)$$

Hence, as long as its price is lower than the reference price of 22.44 €/dt, rapeseed meal is less expensive than any mixture of wheat and soybean meal that provides equal amounts of protein and energy. However, if the price of rapeseed meal is higher than the reference price, it is less expensive to obtain energy and protein from wheat and soybean meal. Table 9.3 presents analogous calculations for a number of substitute feedstuffs.

Table 9.3: Reference prices for selected feedstuffs for dairy cattle compared with wheat and soybean meal

Feedstuff	Raw protein in g / kg	MJ NEL / kg dry matter	Reference price in € / dt
Rapeseed meal	330	6.51	22.44
Molasses pulp	115	6.93	13.09
Grain corn	93	7.38	12.60
Peas	221	7.51	18.59
Barley	109	7.11	13.02

Source: Own depiction based on Groß (2010).

Of course, whether rapeseed meal or a mixture of wheat and soybean meal is fed depends not only on protein and energy, but also on the composition (amino acid profile) of the protein, on other nutrients, and on palatability. In practice, therefore, and depending on the animal species in question, substitution between feedstuffs is only possible within a limited range.

9.2.5. The effect of policy measures at the same processing level

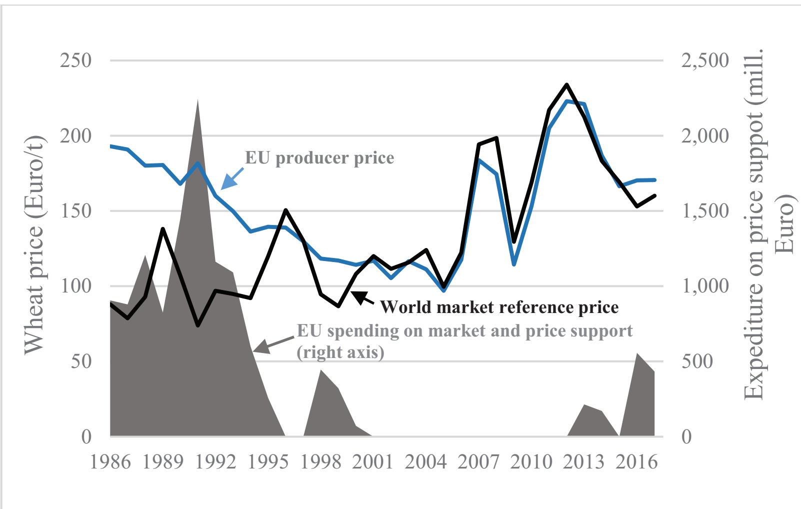
The previous section dealt with the practical importance of technical substitutability for price relationships between feedstuffs. In this section we present two well-known examples from the history of the EU's **Common Agricultural Policy (CAP)** that show how protection for specific domestically produced products was undermined by duty-free imports of substitutes, leading to large economic distortions. The first case (the so-called Open Flank of the CAP) involves cereals and cereal substitutes, the second case involves sugar and alternative sweeteners such as isoglucose.

9.2.5.1. The Open Flank of the CAP

In the CAP, cereals prices are political prices that are influenced by a common market organisation. Under this common market organisation, administrative support prices (so-called threshold and intervention prices) are set by policy makers. In the past, market prices for cereals (and other agricultural products) in the EU were strongly influenced by these administrative prices and the support measures designed to implement them (such as intervention purchases and export subsidies).

In the course of a series of reforms beginning in the early 1990s, the EU's administrative prices were progressively reduced. As a result, producer prices fell, as did the EU's spending on price support measures. Both of these effects can be seen for the case of wheat in Figure 9.1. In addition, and as also apparent in Figure 9.1, starting in 2007/08 world market prices for wheat increased strongly, further reducing the need for price support in the EU.

Figure 9.1: EU producer and world market prices for wheat, and EU expenditure on wheat price support measures



Source: OECD (2022).

The so-called **Open Flank of the CAP** dates back to the beginnings of the CAP, when the EU was designing and first implementing its common market organisations. At this time, the EU (which was then called the European Economic Community – EEC) established a customs union, which means that its members replaced their national tariffs with common EU-wide tariffs. In the early 1960s, before it was clear how high the EU would set its common tariffs, many of the EU's trading partners were worried that it would set high, protectionist and trade-diverting tariffs.[6] As

[6] Josling et al. (1996) describe the negotiations that took place at the time. See Koester (1988) for a detailed discussion of the Open Flank.

a result, in the so-called Dillon Round of GATT[7] negotiations, which lasted from 1960 to 1962, these trading partners – especially the US – demanded compensation from the EU. The EU rejected these demands but did agree to make what appeared, at the time, to be a small concession: it committed itself to import a wide range of oilseeds, protein feeds and cereal substitutes (often collectively referred to as **cereal substitutes**) duty free in perpetuity. Josling et al. (1996, p. 46) consider it "… doubtful if the chief American and EEC agricultural negotiators … realised the significance of their decision at the time it was made". Back then, the EU only imported very small amounts of cereal substitutes. Hence, agreeing to zero or very low levels of import duties for these products appeared, for the European negotiators, to be a harmless concession.

However, in the course of the 1970s this concession developed into a major burden for the CAP because it distorted price relationships between domestic cereals, which were heavily protected, and cereal substitutes, which could enter the EU duty free. Livestock producers in the EU responded to price incentives and increasingly began to feed imported feedstuffs to their animals rather than domestically produced cereals. For example, the right mixture of imported tapioca and soybean meal can largely replace domestically produced wheat or barley. As a result of the Open Flank, total EU imports of cereal substitutes increased from under 2 million tons in the early 1960s to over 18 million tons in 1987 (Josling et al. 1996, p. 46). The cereals that were no longer being fed to animals boosted the EU's cereal surplus and, therefore, its spending on measures such as intervention storage and export subsidies. Imported oilseeds also contributed to growing surpluses and public spending on EU dairy markets. First, margarine produced with oil from duty-free imports of oilseeds reduced the demand for butter in the EU. Second, imported protein feeds replaced skim milk powder in EU feed rations.

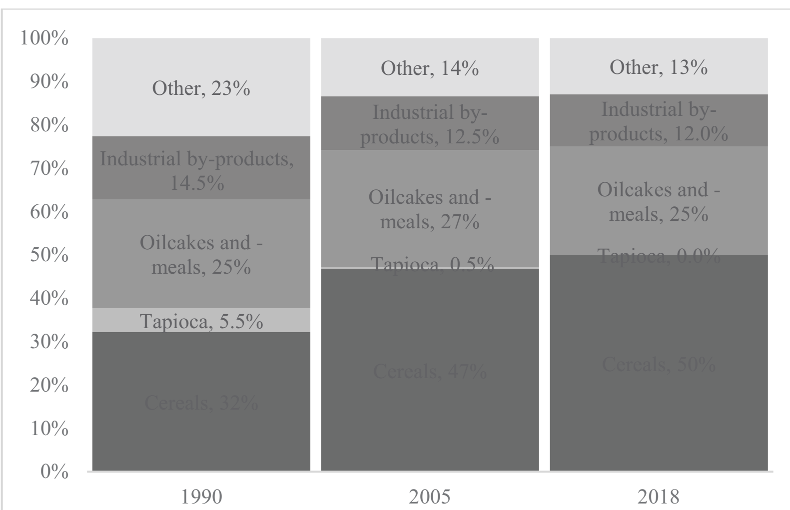
Duty-free imports of cereal substitutes, coupled with price support for domestic cereal and livestock production fuelled a strong expansion of intensive livestock production (beef, pork, poultry and dairy products) in the EU, especially in coastal regions with good access via ports to the world markets for cereal substitutes. The current regional pattern of livestock production in the EU, for example the concentration of German livestock

[7] **GATT** is an abbreviation for *General Agreement on Trade and Tariffs*, the predecessor of the **WTO** (*World Trade Organisation*) in the early 1990s.

production in the north-western Weser-Ems region, is legacy of the Open Flank.

The Open Flank was ultimately closed by the series of reforms referred to above, which progressively reduced the EU's price support for cereals and thus brought EU cereal prices in line with those on world markets (see Figure 9.1). Figure 9.2 shows how the closing of the Open Flank changed the composition of animal feeds in the EU. In 1990, prior to the reforms, tapioca accounted for 5-6% of compound feed in the EU. As cereal prices in the EU fell, tapioca lost its artificial cost advantage. Its share dropped to around 0.5% in 2005, and by 2018 the use of tapioca in animal feed had become negligible. Additional cereal substitutes in the category 'other' experienced similar declines over this period, while the share of cereals grew from one third to one half.

Figure 9.2: The ingredients in compound feed in the EU (1990, 2005 and 2018)



Source: Own depiction with data from FEFAC (2018).

Cereal substitutes are still used in the EU today. Their use depends on their prices relative to cereals, without the cost advantage that existed previously due to the Open Flank. The prices of substitutes relative to cereals, and thus the use of cereals in animal feed, are lower in coastal regions of the EU, and higher in land-locked regions. Nutritional and physiological considerations also influence the composition of animal feed. Soybean meal is a source of high-quality protein (digestibility, amino acid profile). Figure 9.2 shows that the share of oilseed cakes and meals in animal feed in the EU has remained roughly constant (25-27%) over time; imported soybeans (both genetically modified and non-genetically modified) account for the largest share in this category.

9.2.5.2. Sugar, isoglucose and biofuels

Sugar not only plays an important role in human nutrition, it is also an important input in the production of biofuels and in the chemical industry. The term 'sugar' usually refers to sucrose, which is mainly derived from sugar beet and sugar cane. Fructose and lactose are other important types of sugar that are also found in agricultural products, but sucrose is the main output of the sugar industry.

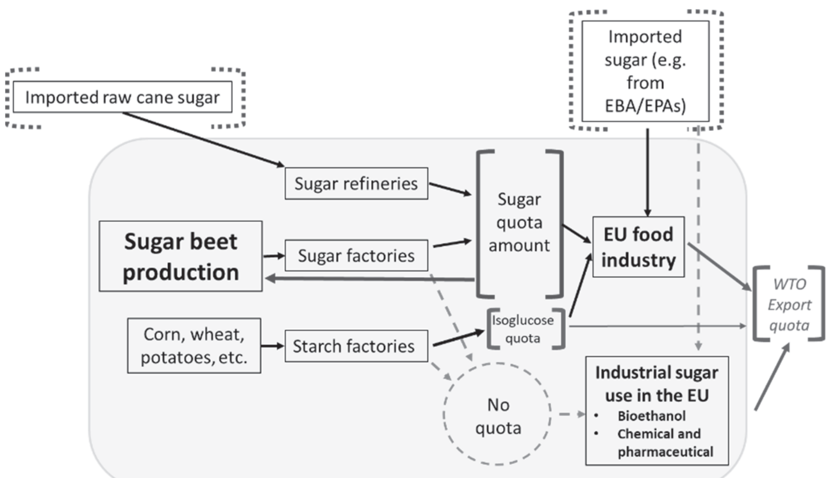
In the EU, there are sugar factories that process sugar beets, as well as refineries that process raw cane sugar. The latter comes from several former French colonies in the Caribbean that remain part of the French state (the so-called Départements d'Outre-Mer), as well as from a number of **African, Caribbean and Pacific** (ACP) countries that have been granted preferential access to the EU sugar market (see below).

The original aim of the EU sugar market organisation was to support the sugar price by restricting sugar beet production. To this end, a fixed production quota was established. This quota was distributed among the sugar processing firms, which further allocated it to individual sugar beet producing farms in the EU. By artificially reducing supply, the quota increased sugar prices in the EU. To keep out imports of cane sugar that would otherwise have been attracted by high prices and thus undermined the quota, the EU also introduced high import duties for sugar. However, it granted some low-income ACP countries duty-free access to the EU market under schemes such the EPA (**Economic Partnership Agreements**) and the EBA (**Everything but Arms**) initiative. Despite the quota, the EU remained over 100% self-sufficient in sugar, and the preferential imports added to its export surplus. In 2005, a WTO panel found that the EU was exporting more sugar than permitted with the help of export subsidies.[8] As a result of this finding, the EU was obliged to reduce its sugar exports.

[8] The EU argued that it was not subsidising its sugar exports at all. However, the panel found that sugar companies were using their inflated earnings on the domestic EU market to indirectly subsidise exports. See Burrell et al. (2014) for details.

Figure 9.3 summarises the complex constellation of interacting markets, stakeholder interests and international commitments that EU policy makers had to navigate in their efforts to support prices for domestic sugar beet producers. The oval background area represents the EU internal market that was protected by external tariffs on sugar imports, but to which some countries had preferential access (dotted brackets at the top). Rectangular brackets symbolise quantity restrictions. Food industry use and industrial sugar use formed two separate market segments until the sugar quote was abolished on October 1, 2017.

Figure 9.3: Schematic depiction of substitution relationships and interactions on the EU sugar market from 2006 to the end of the quota system in 2017



Source: Own depiction based on Agrosynergie (2011).

Sugar can be largely and sometimes completely replaced by glucose-fructose syrup – which is often referred to as isoglucose – in a number of food industry uses, for example to sweeten soft-drinks. Isoglucose is not produced from sugar beets or sugar cane, but rather from the starch in products such as corn, wheat, and potatoes. If the EU sugar market organisation had only imposed a production quota on sugar from beets and cane, making it more expensive, then the food industry would have replaced as much sugar as possible with isoglucose, triggering an expansion in isoglucose production. This would have undermined the sugar quota and depressed sugar prices. To avoid this, EU agricultural policy makers had to impose a quota on isoglucose production as well.

Figure 9.3 also shows that sugar and starch for industrial use were not subject to any quantitative restrictions. For example, there was no limit on the production of biofuels from sugar. Hence, if fossil fuel prices were high enough, ethanol production based on unrestricted 'industrial' sugar beets became attractive.

Since the sugar quota was abolished on October 1, 2017, prices for domestic beet sugar, imported cane sugar, and isoglucose are freely determined. Nevertheless, the linkages between the beet sugar, cane sugar, starch and isoglucose markets depicted in Figure 9.3 remain. These linkages connect food markets to industrial and fuel markets, resulting in a complex, dynamic price structure.

9.3. Price relationships between joint agricultural products

In the previous section we considered price relationships between agricultural products at the same processing stage, for example different feedstuffs that compete for inclusion in feed rations. In this section we turn our attention to the fact that many agricultural products contain several components that can be processed into different intermediate or final food products. For example, oilseeds such as soybeans and rapeseed can be processed into oil and protein (meal) components by crushing. The resulting **joint products** are sold for further processing and/or final consumption on different markets, where they do not compete with one another.[9] Similarly, higher- and lower-quality cuts of meat from, for example, a chicken, can be viewed as joint products, although in this case they do compete because consumers might switch from higher- to lower-quality cuts if prices increase, and vice versa. A further example of joint production is milk, which has the two main components fat and protein. Dairy products contain different combinations of these two components: for example, butter is mainly composed of the fat component, while skim milk powder is mainly composed of the protein component. Note that not all joint products are separated from one another at the processing stage, like cuts of meat in a slaughterhouse or fat and protein in a dairy processing plant. For example, milk and meat emerge separately as joint outputs of dairy farming.

[9] The relationship between prices for agricultural products at different processing stages, for example the relationship between the prices of soybeans and soybean oil and meal, are referred to as vertical price relationships, and are discussed in detail in Chapter 10.

9.3.1. The supply of joint products[10]

In section 9.2.1 we discussed the connection between two markets that arises when one (e.g. grain) is an input in the production of another (e.g. poultry meat). The opposite is the case if two outputs are generated by a production process, and (some of) the inputs into this process cannot be unambiguously assigned to one or the other output. Most agricultural production processes produce multiple outputs using multiple inputs, and often it is not possible to clearly allocate all of the input use to specific outputs, i.e. to measure what amount of an input Z is responsible for output Q_1, and what amount is responsible for another output Q_2.

The implicit production function in equation (20) describes a production system that produces two joint outputs, Q_1 and Q_2, using two inputs, Q^x and Z.

$$F(\mathrm{Q}_1, \mathrm{Q}_2, Q^x, \mathrm{Z}) = 0 \tag{20}$$

The input Q^x can be allocated, i.e. we know how much of it led to the production of Q_1 (Q_2). The other input Z cannot be allocated. We can re-write the production function as follows, where $Q^x = Q_1^x + Q_1^x$:

$$Q_1 = f^1(Q_1^x, Z) \tag{21}$$

$$Q_2 = f^2(Q_2^x, Z) \tag{22}$$

A typical example of a production process according to equation (20) is the production of milk Q_1 and beef Q_2 using feed Q^x and a cow Z. The feed use can be allocated to the individual outputs using information on the underlying physiological relationships, but the cow, which produces milk and calves jointly, cannot be allocated.

From the perspective of profit maximisation, the production function in equation (20) leads to the following optimisation problem:

$$\pi = P_1 Q_1 + P_2 Q_2 - P^x Q_1^x - P^x Q_2^x - P^z Z \tag{23}$$

or

$$\pi = P_1 f^1(Q_1^x, Z) + P_2 f^2(Q_2^x, Z) - P^x Q_1^x - P^x Q_2^x - P^z Z \tag{24}$$

[10] This section is based on Chapter 5.5 in Beattie et al. (2009).

The first-order conditions for a profit maximum are determined by differentiating equation (24) with respect to the input quantities and setting the derivative equal to zero:

$$\frac{\partial \pi}{\partial Q_1^x} = P_1 \frac{\partial Q_1}{\partial Q_1^x} - P^x = 0 \quad (25)$$

$$\frac{\partial \pi}{\partial Q_2^x} = P_2 \frac{\partial Q_2}{\partial Q_2^x} - P^x = 0 \quad (26)$$

$$\frac{\partial \pi}{\partial Z} = P_1 \frac{\partial Q_1}{\partial Z} + P_2 \frac{\partial Q_2}{\partial Z} - P^z = 0 \quad (27)$$

Equations (25) and (26) tell us that profit is maximised when the price of the input that can be allocated (e.g. feed) is equal to its marginal value product in the production of Q_1 (e.g. milk) and to its marginal value product in the production of Q_2 (e.g. beef). Hence, these two marginal value products must be equal. Rearranging equation (27) to isolate P^z on the right-hand side reveals a further condition for profit maximisation: that the price of the input that cannot be allocated (e.g. the cow) equals the sum of its marginal value products for Q_1 and Q_2 (e.g. milk and beef).

By simultaneously solving equations (25), (26) and (27) for the input quantities Q_1^x, Q_2^x and Z, we obtain the (uncompensated) input demand functions.[11] These demand functions are referred to as 'uncompensated' because they capture both substitution effects and effects due to changes in the output quantities that result from price changes.[12] These functions show that the demand for an individual input is not only a function of its own price, but also of the two output prices and the price of the other input. Furthermore, inserting the input demand functions into the production functions in equations (21) and (22), we obtain the output supply functions which show that the supply of each output is also a function of all input and output prices. This means, for example, that the output of beef depends not only on the beef price and the feed price, but also on the milk price, since beef and milk are joint products that are linked by the input that cannot be allocated (in this example, the cow).

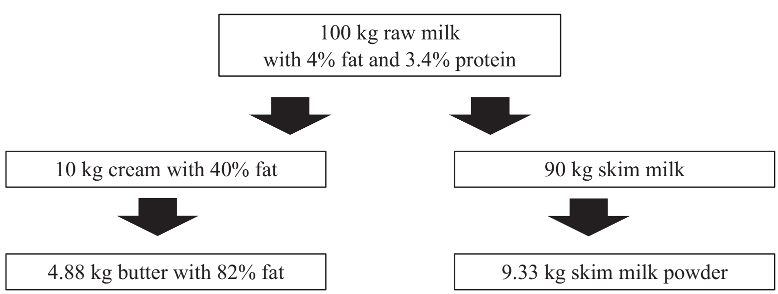
[11] For the derivation of these input demand functions and the output supply functions referred to below see Beattie et al. (2009).
[12] Compensated or Hicksian input demand functions compensate or correct for output changes to isolate substitution effects. For details see Beattie et al. (2009) and Snyder et al. (2012).

Whether or not an input can be allocated to different outputs is somewhat arbitrary and depends on the costs of a detailed, definitive analysis. It might be possible, based on detailed analysis of physiological and metabolic processes, to trace and separate a cow's contributions to milk and meat production. However, this would not change the fact that in reality the living cow is a single input with a single market price.

9.3.2. Price relationships between joint products – the example of milk

In this section we take a closer look at price relationships between joint products by analysing the case of raw milk that is processed into butter and skim milk powder (SMP). This analysis can be applied to other markets, for example oilseeds that are processed into oil and meal. Figure 9.4 presents several important technical relationships that apply in the case of milk.

Figure 9.4: Technical relationships between milk products at different processing levels*



Source: Own depiction based on DairyCo (2014).
** Simplified, and without considering processing losses.*

Figure 9.4 shows that the main marketable components of raw milk are the fat component and the protein component. The market demand for raw milk is therefore ultimately derived from the final consumer demand for products that contain milk fat on the one hand, and for products that contain milk protein on the other. In the following we simplify by assuming that the fat component can only be processed into butter, and the protein component only into SMP.

If only the fat component of milk had a market value, we could derive the price of raw milk from the price of butter. We would start with the butter price, subtract processing and marketing costs, and divide by an appropriate **technical transformation coefficient** (see Figure 9.4) that indicates how many kg of milk are required to produce 1 kg of butter. However, dairies can also process and sell the protein component of milk. Hence, the price that they can pay farmers for raw milk will also reflect the value of the protein component, as reflected in the following equation:

$$MV^{raw\ milk} = bP^{butter} - bc^{butter} + dP^{SMP} - dc^{SMP} \qquad (28)$$

In equation (28), the marginal value of one kg of raw milk ($MV^{Raw\ milk}$) equals the revenue from the sale of its fat and protein components (bP^{butter} and dP^{SMP}, respectively), minus the costs of processing each of these components (bc^{butter} and dc^{SMP} , respectively). Table 9.4 presents information on these costs, and based this information Table 9.5 calculates the marginal value of one kg of raw milk using market prices for butter and SMP in the EU in June 2018.

Table 9.4: Assumed costs of processing 100kg of raw milk into butter and skim milk powder (SMP)

Symbol in equation (28)	Description	Source / Assumption
MV	Derived marginal value of raw milk	Calculated using equation (28)
P	Market prices	Current market prices
b	Technical transformation coefficient for butter	100 kg raw milk → 4.88 kg butter, or 20.49 kg raw milk → 1 kg butter (see Figure 9.4)
c	Processing and marketing costs*	€ 32.03 / 100 kg butter € 43.47 / 100 kg SMP
d	Technical transformation coefficient for SMP	100 kg raw milk → 9.33 kg SMP, or 10.72 kg raw milk → 1 kg SMP (see Figure 9.4)

Source: Own depiction based on DairyCo (2014)
** Processing costs do not include the cost of the raw milk itself.*

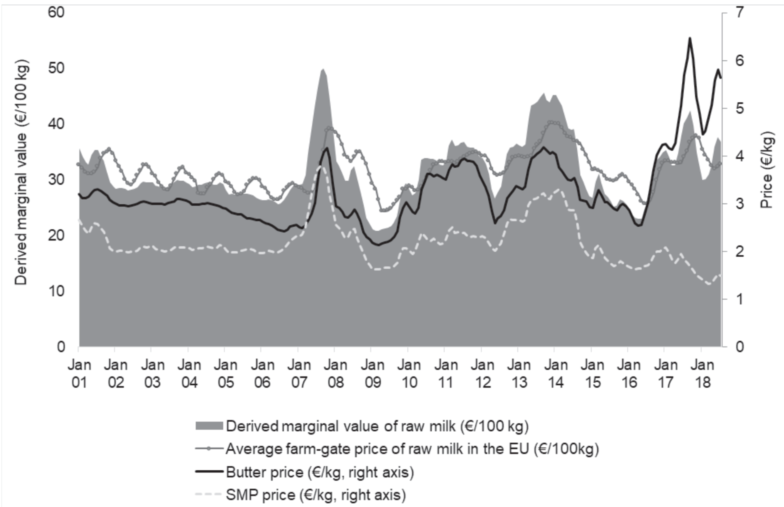
Table 9.5: The marginal value of raw milk in the EU in June 2018 based on average butter and skim milk powder (SMP) prices

Butter price (€/kg)	5.82
Costs of processing for butter (€/kg)	0.32
SMP price (€/kg)	1.52
Costs of processing for SMP (€/kg)	0.43
Derived marginal value of raw milk (€/kg)	**0.37**

Source: Own illustration based on equation (28) and Table 9.4.

The marginal value of raw milk derived according to equation (28), and illustrated in Table 9.5, is based on a number of assumptions. Do dairy processors pay corresponding prices to farmers for the raw milk that they deliver? Figure 9.5 shows that the average monthly farm-gate price for raw milk in the EU is correlated with the derived marginal value, but not perfectly. Since 2007 in particular we can see that movements in the farm-gate price have tended to lag behind movements in the derived marginal value by several months. We can also see that the farm-gate price fluctuates less (lower peaks and higher troughs) than the derived marginal value.

Figure 9.5: Derived marginal values of raw milk and average farm-gate prices of milk in the EU (monthly averages, 2001-2018)



Source: Own depiction with data from the EU Commission (2020b).

Possible explanations for these observations include the following:

- Dairy processors try to smooth the prices that they pay to farmers for raw milk over time.

- The actual marginal value of a kg of raw milk does not depend exclusively on butter and SMP prices. Diary companies process milk into a much wider range of products, including for example cheese. Cheese contains both fat and protein, and some varieties of cheese command premium prices on consumer markets. The mix of consumer products that a dairy company produces has a strong influence on the prices that it can pay farmers for raw milk; companies that produce more high-value products such as cheese can pay farmers higher prices for raw milk.

9.3.3. The effects of policy measures on prices for joint products

In this section we discuss several examples of how political market interventions influence and are influenced by price interactions between joint products.

9.3.3.1. The EU's intervention price system for milk

The EU's CAP attempts to support the farm-gate price of raw milk by setting support prices for butter (the fat component of milk) and SMP (the protein component) (see the previous section). This makes sense, because raw milk is perishable and expensive to store even in the short run. Butter also requires refrigeration, but it is less perishable and has lower storage costs relative to its value. Hence, so-called intervention purchases of butter and SMP are used to indirectly support the price of raw milk in the EU. Support policy could focus instead on other processed dairy products such as cheese, but butter and SMP have the advantage of being comparatively homogeneous.

As a result of reductions in support prices and increases in world market prices for cereals, the EU has not intervened significantly on cereals markets since roughly the turn of the century (see Figure 9.1). Cereal support prices still exist, but market prices have been consistently and often considerably higher. The situation on EU milk markets is different, as support prices and the so-called intervention system have continued to play an important role through a series of milk price crises.

The farm-gate price of milk is not directly supported by the intervention system; instead this system supports the prices of butter and SMP for dairy companies and thus puts them in a position to pay higher prices to the farmers who deliver milk. For example, Table 9.6 presents calculations – similar to those in Table 9.5 above, but more detailed – that show that in 2015, EU intervention prices for butter and SMP supported a derived farm-gate price for raw milk of 19.7 Cent/kg.

Table 9.6: Derivation of the farm-gate value of raw milk from intervention prices for butter and skim milk powder (SMP)

	Item	Value	Unit
	Reference price butter	246.4	€/100 kg
I	Intervention price butter (90% of reference)	221.8	€/100 kg
II	Reference price SMP	169.8	€/100 kg
	Transport costs butter/SMP		
	Dairy to intervention warehouse	350.0	km
IIIa	- Transport costs butter	29.8	€/t
IIIb	- Transport costs SMP	23.5	€/t
	Financial costs		
IVa	- Financial costs butter	23.7	€/t
IVb	- Financial costs SMP	18.1	€/t
	Marketing and processing costs		
V	- Marketing costs	10.0	€/t
VI	- Processing costs butter/SMP	320.0	€/t

Value of raw milk to dairy processor derived from intervention prices:

$$\underbrace{(\mathrm{I} - (\mathrm{IIIa}+\mathrm{IVa}+\mathrm{V}+\mathrm{VI})/20.49)}_{\text{Fat component (butter)}} + \underbrace{(\mathrm{II} - (\mathrm{IIIb}+\mathrm{IVb}+\mathrm{V}+\mathrm{VI})/10.72)}_{\text{Protein component (SMP)}}$$

= Value of raw milk at processing plant	21.1	Cent/kg
- Collection costs from farm to plant	1.44	Cent/kg
Farm-gate value of raw milk	19.7	Cent/kg

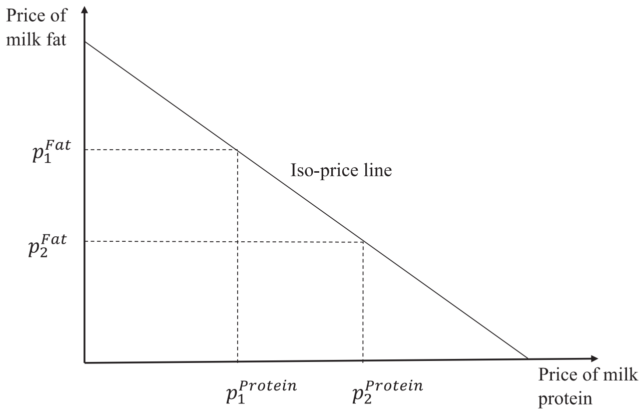
Source: Thiele et al. (2015). Calculations assume milk with 4% fat and 3.4% protein. Financial costs are caused by delays in receiving payment for butter and SMP delivered to intervention. See Thiele et al. (2015) for details.

Whether dairy companies in the EU pass on supported prices for butter and SMP to milk-producing farmers depends on several factors. As mentioned above, dairy companies do not only produce butter and SMP. Some produce higher-value products that can be sold at premium prices on consumer markets, and are thus able to pay farmers more for raw milk. In addition, since the dairy industry is highly concentrated in Germany[13] and many other EU countries[14], dairy companies may be able to exercise market power to depress the prices that they pay farmers for raw milk. In 2016, for example, during a period of relatively high milk production (due in part to abolishment of the milk quota in 2015), spot market prices for raw milk fell significantly below the values derived from intervention prices for butter and SMP.

Figure 9.6 shows how the value of raw milk can be derived from the prices of its fat and protein components. The **iso-price line** for milk in Figure 9.6 is the set of all combinations of the prices for the fat component and the protein component that lead to the same value of raw milk. It follows that the relationship between the price of raw milk and the price of the processed product butter is not constant, but rather depends on the price that is obtained for the protein component. A given value of raw milk can be attained with high fat and low protein prices, but also with low fat and high protein prices. The slope of the iso-price line is determined by the relative amounts of fat and protein in a kg of raw milk. Which point on the iso-price line is realised depends on demand conditions for different milk products and the resulting relative prices of fat and protein.

[13] Bundeskartellamt (2012).
[14] EU Commission (2016).

Figure 9.6: Iso-price line for raw milk



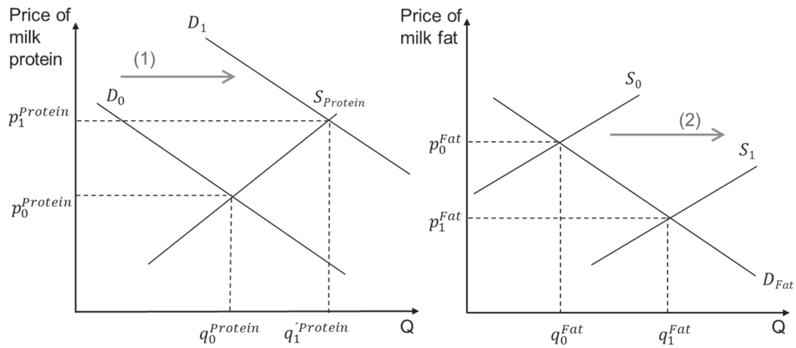
Source: Own depiction.

In the following it will be shown how the price relation of protein to fat changes as a result of shock to the equilibrium in one of the markets:

The intervention prices of butter and SMP will, by influencing the prices of milk fat and milk protein, also influence the prices of other milk products that are not subject to intervention. For example, as a result of an increase in the intervention price for butter, less milk fat will be available for the production of other dairy products that require milk fat, such as some types of cheese. The increase in the intervention price for butter increases the opportunity costs of using milk fat in other dairy products. This shifts the supply curve of these products upward, and increases their prices. These price increases will be smaller than the original increase in the price of butter, unless the demand for these other products is perfectly inelastic, or their supply is perfectly elastic. Hence, while the market prices of butter and SMP will follow the corresponding intervention prices closely, the prices of other dairy products that contain milk fat will be affected less strongly, depending on product-specific market conditions.

Imagine a situation in which the demand for milk protein increases, i.e. the demand curve shifts to the right (from D_0 to D_1 in the left panel of Figure 9.7). This might occur as a result of an increase in consumer demand for 'high protein' foods that are produced using milk protein. The price of milk protein will increase and with it the price of milk. This will trigger an increase in milk production, leading to an increase in the quantity of milk protein supplied (from $q_0^{Protein}$ to $q_1^{Protein}$).

Figure 9.7: The effect of an increase in the demand for milk protein on the markets for milk protein and milk fat



Source: Own depiction.

However, since milk protein and milk fat are joint products that are produced in fixed proportion[15], the increase in protein production will be accompanied by an increase in fat production as the supply curve of milk fat shifts to the right (from S_0 to S_1 in the right panel of Figure 9.7). The original increase in demand for milk protein therefore leads to two opposite price movements: an increase in the price of milk protein, and a reduction in the price of milk fat. The result is a shift in the price relationships among raw milk and its fat and protein components.

9.3.4. Energy prices and agriculture

The interaction between agricultural prices and energy prices has become increasingly important in recent years. In this section we will discuss three aspects of these interactions. These three aspects do not cover all dimensions of the interactions between agriculture and energy markets, but they illustrate how agricultural price relationships can be influenced by policies and market interventions in other, non-agricultural sectors of the economy.

[15] This proportion is fixed in the short run. However, farmers can influence the composition of milk by changing the cows' feed, and in the longer run by switching to other breeds of dairy cow.

9.3.4.1. Food or fuel: agricultural prices and the energy market

The finite nature of fossil fuel supplies and the problematic accumulation of CO_2 from non-renewable sources in the atmosphere has prompted many governments worldwide to subsidise the production of energy using agricultural raw materials, such as ethanol based on sugar, and biogas (methane) based on corn silage. The aim of these subsidy programs is usually to improve the competitiveness of energy (such as electricity or fuels) from renewable sources vis-à-vis traditional, non-renewable fossil energies.

These policies have given farmers in many countries new options for generating income with their land, labour, and capital. At the same time, however, these policies have highlighted the fact that bioenergy production competes with food production for limited agricultural land – the so-called **food or fuel conflict**. Awareness of this conflict was heightened when world market prices for agricultural raw materials increased sharply in 2006/07.[16]

For example, since 2009 the EU's Renewable Energy Directive (RED) requires that 10% of the fuel used in European road transport must come from renewable sources. The new edition of this directive (RED II) requires that in all EU member states 14% of the energy used in road and rail traffic must come from renewable sources by 2030 (EU Commission 2019). This has led to significant production of biodiesel using rapeseed in the EU; ethanol production based on sugar beet, wheat or corn plays a smaller role. US biofuel policy has promoted ethanol production from corn and thus influenced world market prices for these products (Koirala et al., 2015), while in Brazil the government's *Proálcool* program has supported the production of ethanol from sugar cane since 1975 (Andrietta et al. 2007).

The EU's promotion of biodiesel production is a controversial issue because it affects oilseed prices, but also because it has led to increased imports of palm oil for biodiesel production. New technologies that make palm oil more attractive relative to rapeseed have contributed to this trend (Bureau and Swinnen, 2018). This in turn has contributed to the expansion of palm oil plantations in Asia but increasingly also in South America and Africa, which can have negative effects on biodiversity and the maintenance of rainforests.

[16] See Mitchel (2008).

At the same time, EU and US biofuels policies have also created employment and income opportunities in agriculture in many low-income countries. These can contribute to economic development and improvements in food security. Households that are net producers of agricultural products will generally benefit from higher demand and prices for agricultural products as a result of policies that encourage the production of biofuels. However, low-income households that spend a significant proportion of their income on food will be negatively affected by global food price increases (Swinnen 2011).

9.3.4.2. Competition for land

Renewable energy production can compete with food production even if the relevant production factors, land and capital, are not exclusively owned by farmers. Food and energy production compete with one another for the use of land. The resulting changes in agricultural factor prices will have an indirect effect on the prices of agricultural products.

As an example of this consider the influence of wind turbines on agricultural land prices in Germany. In an attempt to increase the share of renewable energy in national energy use, many European countries, including Germany, have subsidised wind turbines that generate electricity. In Germany this is accomplished by means of the EEG (Renewable Energy Law) which obliges utility companies to purchase electricity from wind turbines at guaranteed prices, and to pass on the resulting additional costs to final consumers of electricity.

Locations that have frequent and sustained winds are especially attractive for the construction of wind turbines. Most wind turbines are constructed on agricultural land, but even the largest models do not require much area. Compared with other renewable energies such as biodiesel or biogas, therefore, electricity generation with wind turbines does not interact directly with agricultural production.

However, since the EEG makes it profitable to operate a wind turbine, and since these turbines cannot be built too close to one another, operators compete for spaces and pay correspondingly high rental or purchase prices to landowners at favoured locations. Ritter et al. (2015) present evidence that agricultural land prices have increased strongly at such locations.

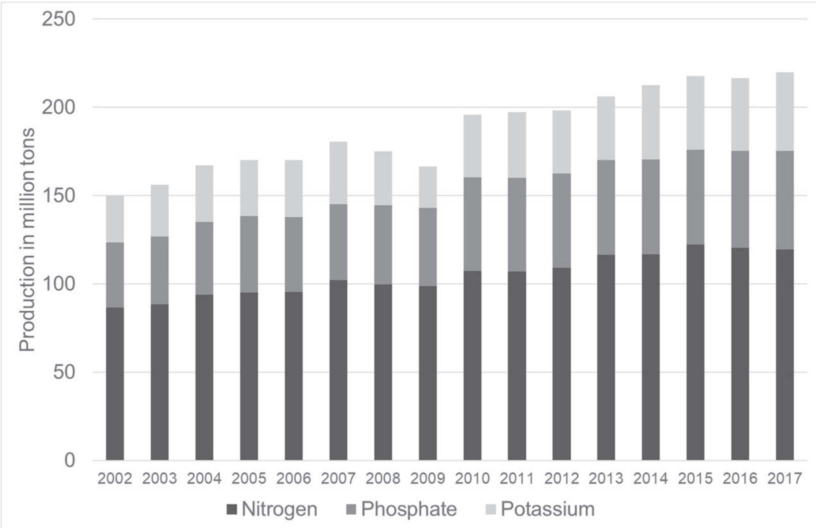
The increasing generation of renewable energy can thus increase competition for land and, by increasing land prices, also affect the competitiveness of agricultural production in some locations. Similar to the case of wind turbines in Germany, extensive solar parks in other parts of the world can also compete with agricultural production for land.

9.3.4.3. Energy prices and the marginal cost of agricultural production

Energy is generally not considered to be one of the primary factors of production in agriculture (land, labour and capital). Nevertheless, many agricultural production processes use energy – from renewable or non-renewable sources – for heating, drying or to power machinery. Furthermore, as outlined above, energy prices can indirectly affect land prices. Energy costs therefore account for an important share of production costs for many types of agricultural production.

Energy prices also affect agriculture via fertiliser costs. Figure 9.8 shows that the global production of mineral fertilisers continues to increase steadily. The production of nitrogen fertilisers using the **Haber-Bosch process** is particularly energy-intensive. Altogether, fertiliser production accounted for roughly 2% of global energy use in the 1990s, and the Haber-Bosch process accounted for 5% of global natural gas use (IFA 1998, p. 33-34). Hence, the influence of fertiliser production on energy prices is likely limited. Nevertheless, the production and prices especially of nitrogen fertiliser depend heavily on energy prices.

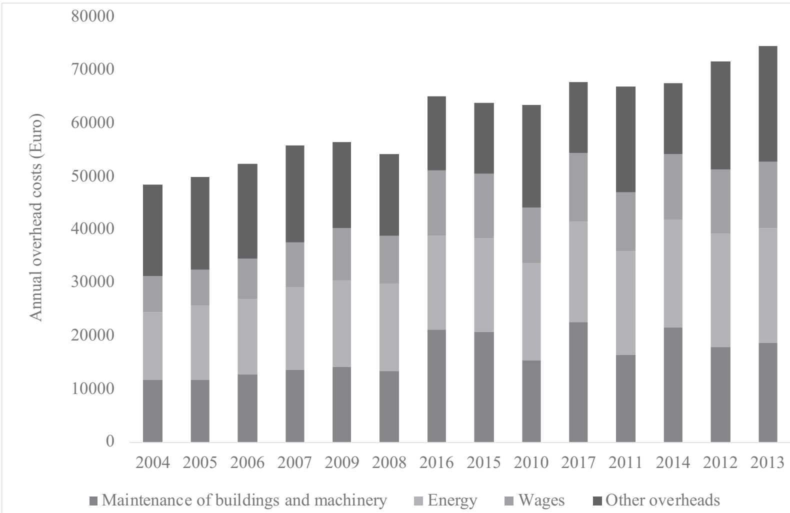
Figure 9.8: Global fertiliser production



Source: Own depiction with data from FAO (2020).

Because it is difficult to allocate spending on energy to different types of production on a farm, it is often considered among overhead costs. Figure 9.9 presents information on the composition of overhead costs in German agriculture from 2004 to 2013. We can see that over time the energy costs developed approximately in proportion to the other overhead costs and accounted for between 25% and 30% of the total. Figure 9.9 is based on data from the FADN (Farm Accountancy Data Network), which the EU operates in all member states (EU Commission 2020a). It depicts average values across all farm types and size classes in all German regions. Clearly, such averages hide considerable variation, and the share of energy costs can vary considerably from farm to farm and over time.

Figure 9.9: The share of energy costs in total overhead costs of agricultural holdings in Germany



Source: Own depiction with FADN data (EU Commission 2020a).

The share of energy in the costs of transporting and processing food vary considerably. In some energy-intensive cases, for example when products must be refrigerated or frozen, or when SMP is produced by drying, energy costs can play a very important role.

These considerations show that changes in energy prices on world markets, but also changes due to government interventions on domestic energy markets, can influence the competitiveness of individual agricultural products, and the competitiveness of agriculture relative to other sectors of the economy. In the future, trading CO_2 emission certificates could create a new source of income for agriculture. In comparison with other energy-intensive sectors, some types of farming activity can create CO_2 sinks, for example by increasing the humus content of soils or by means of bioenergy generation with carbon capture and storage (BECCS). Not all farms will be able to take advantage of trade in emission certificates to the same extent, and the benefits will depend on whether certificate prises increase in line with energy prices in the future (see Sands et al. 2011).

9.4. Keywords

- ACP (African, Caribbean and Pacific)
- Agricultural price structure, price relationships, price ratios
- Common Agricultura Policy (CAP)
- Cereal substitutes
- Dual cost function
- EBA (Everything But Arms)
- EPA (Economic Partnership Agreements)
- Food or fuel conflict
- GATT (General Agreement on Trade and Tariffs)
- Haber-Bosch process
- Hedonic pricing model
- Iso-price line
- Joint products
- Löhr’s method
- The Open Flank of the CAP
- Pivotal agricultural price
- Production elasticity
- Shephard's lemma
- Technical transformation coefficient
- WTO (World Trade Organisation)

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Exercises

1. In 2017, butter prices in the EU reached a very high level. Nevertheless, butter production initially increased very little. Could this be explained by the fact that cheese prices were also relatively high at the same time?

2. In the EU, many supermarket chains and some dairies required that farmers only feed non-GM soy to their cows. Discuss what effect this will have on the supply and demand of domestic protein crops.

3. The pork producers in a closed economy produce according to the following function $Output_{Pork} = q_{Labour}^{0.2} q_{Feed}^{0.8}$. If the market price for pork is $Price_{Pork}$ = 20 €/dt, how high are the wages and feed prices per unit of labour and feed, respectively?

4. Assume that after abolishing the milk quota, the EU would like to slow down the increase in milk production by raising beef prices. Discuss the short-run and the long-run effects of this measure.

5. In the function $c^{min} = 2Qw_G^{0.5}w_E^{0.5}$, production costs are a function of the output quantity Q and the input prices w_G and w_E, but input quantities are not mentioned. Explain why this is the case.

6. Assume that the wheat price in the EU is 15 €/dt, and that the price for non-genetically modified soybean meal increases to 40 €/dt. Using Löhr's method and the information in Table 9.3, calculate the maximum market prices up to which European peas and grain corn can compete in feed rations.

7. By lowering its domestic grain prices, the EU has reduced domestic demand for imported cereal substitutes such as oilseeds. Discuss the effects of this policy change on world market prices for vegetable oils.

8. Data Exercise: Search the internet for estimates of price elasticities of demand for different oilseed products in the EU and China. A possible source is USDA-ERS (2019). Both the EU and China are important importers of oilseeds on the world market. How do you interpret their respective elasticities?

9. Data exercise: Download recent price data from the *Milk Market Observatory* (EU Commission 2020b) and use equation (28) as well as the information in Table 9.4 to calculate the marginal value of raw milk. Compare this with the average EU farm-gate price for raw milk.